



DESIGN THINKING PROJECT

GROUP - 11

HydraSense – Seeing the Invisible in Rural Drinking Water

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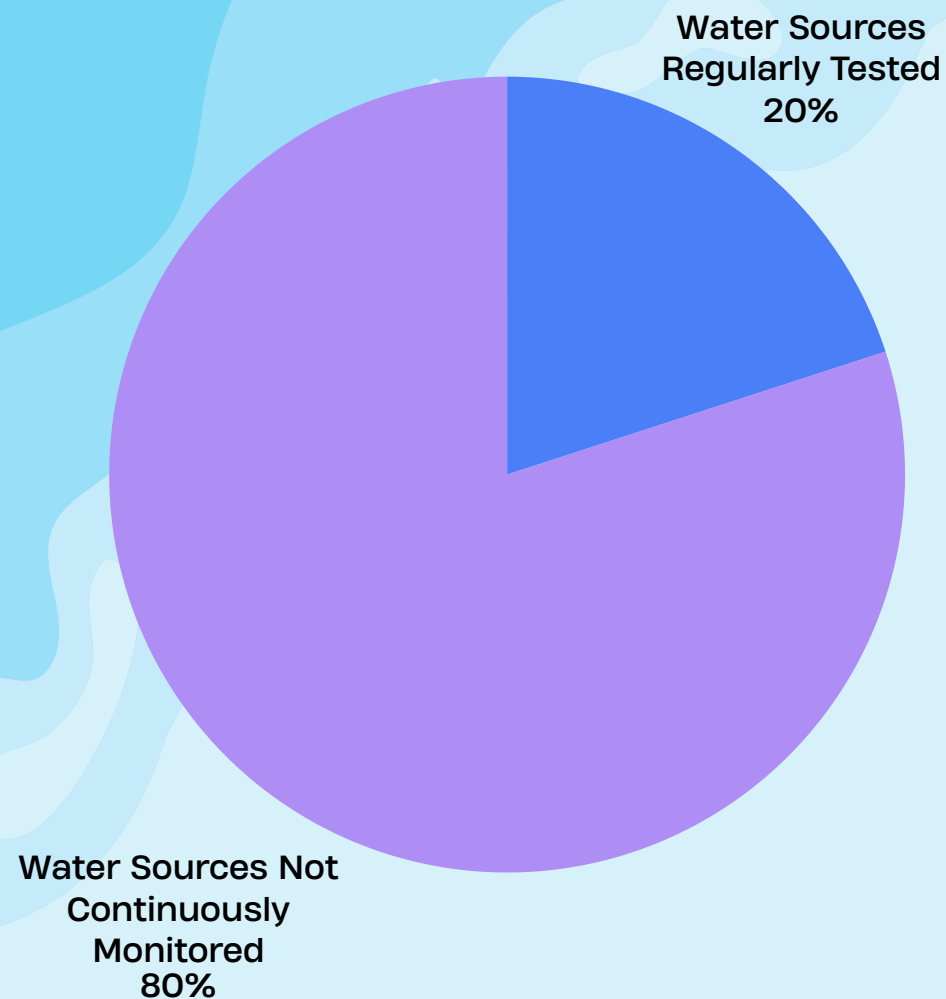
Problem Statement



- Despite major investments under Jal Jeevan Mission, many rural communities still do not have reliable access to safe drinking water.
- Water contamination is often invisible to human senses due to contaminants like **bacteria, excess chlorine, and high turbidity**, making it difficult for villagers to determine whether water is safe for consumption.
- Existing water monitoring methods are either manual, expensive, or not capable of providing real-time alerts to rural users.
- Water distribution within rural pipeline networks is often unequal, where households at the tail end receive less water than those near the source.
- There is a need for a **low-cost, solar-powered, IoT-based monitoring system** that can continuously track water quality and distribution conditions.



Why this matters ?



- 2 Billion+ people globally lack safely managed drinking water
- 106 million people continue to rely directly on untreated surface water sources such as rivers, ponds, and lakes.
- Rural communities remain disproportionately affected, with safely managed drinking water coverage only 60% in rural areas compared to 83% in urban areas
- Waterborne diseases cause thousands of preventable illnesses every day
- Rural communities need affordable real-time monitoring solutions
- The pie chart at the left shows the prevalent Rural Water Monitoring Gap



Core challenges in rural areas



1) Technical Gap

No affordable, low-power IoT sensors exist for continuous rural water quality monitoring.

2) Infrastructure Decay

Aging pipelines and distribution networks fail without predictive maintenance or leak detection.

3) Social Awareness

Villagers lack real-time visibility into water safety, preventing timely community action.

4) Economic Constraints

High-cost SCADA systems are unaffordable for village-level deployment and maintenance (By Gram Panchayats).

Cost of Contamination: Real-World Water Failures



Case Study : Indore Drinking Water Contamination Crisis (December 2025)

Bhagirathpura (Indore) Water Crisis • Dec 2025

- **The Crisis:** Major municipal drinking water contamination in Bhagirathpura, Indore.
- **Root Cause:** Sewage leaked into the drinking water network through damaged pipelines.
- **Delayed Detection:** Contamination went unnoticed for days, continuously supplying toxic water.
- **The Toll:** 1,400+ residents affected, 200+ hospitalized, and 32 deaths.
- **Systemic Failures:** Broken infrastructure and a total lack of continuous water quality monitoring.



Case Study: Nongpok Sekmai Rural Water Crisis (2024)

Nongpok Sekmai Water Crisis (Manipur) • 2024

- **The Crisis:** Severe rural water crisis persisted despite safe-water initiatives.
- **Root Cause:** Heavy reliance on contaminated open wells and deteriorating rivers due to a lack of treated water infrastructure.
- **Environmental Strain:** Illegal sand and gravel mining reduced river flow and severely degraded water quality.
- **Visible Pollution:** Water sources suffered from high turbidity, extensive algae blooms, and floating debris.
- **Human Toll:** Rising waterborne illnesses and high financial burdens on families forced to buy clean water



Our Solution

Solar-Powered Smart Rural Water Monitoring System

A low-cost, solar-powered water monitoring system that continuously measures water quality parameters and provides timely alerts when unsafe conditions are detected.

Problems solved:

- Monitors pH level, TDS and turbidity in real time.
- Conditions and converts sensor outputs into digital data.
- Processes measurements using an MSP430 microcontroller.
- Transmits data wirelessly to a central dashboard or user interface.
- Generates alerts when water quality exceeds safe limits.
- Operates using solar energy for sustainable deployment in rural areas.

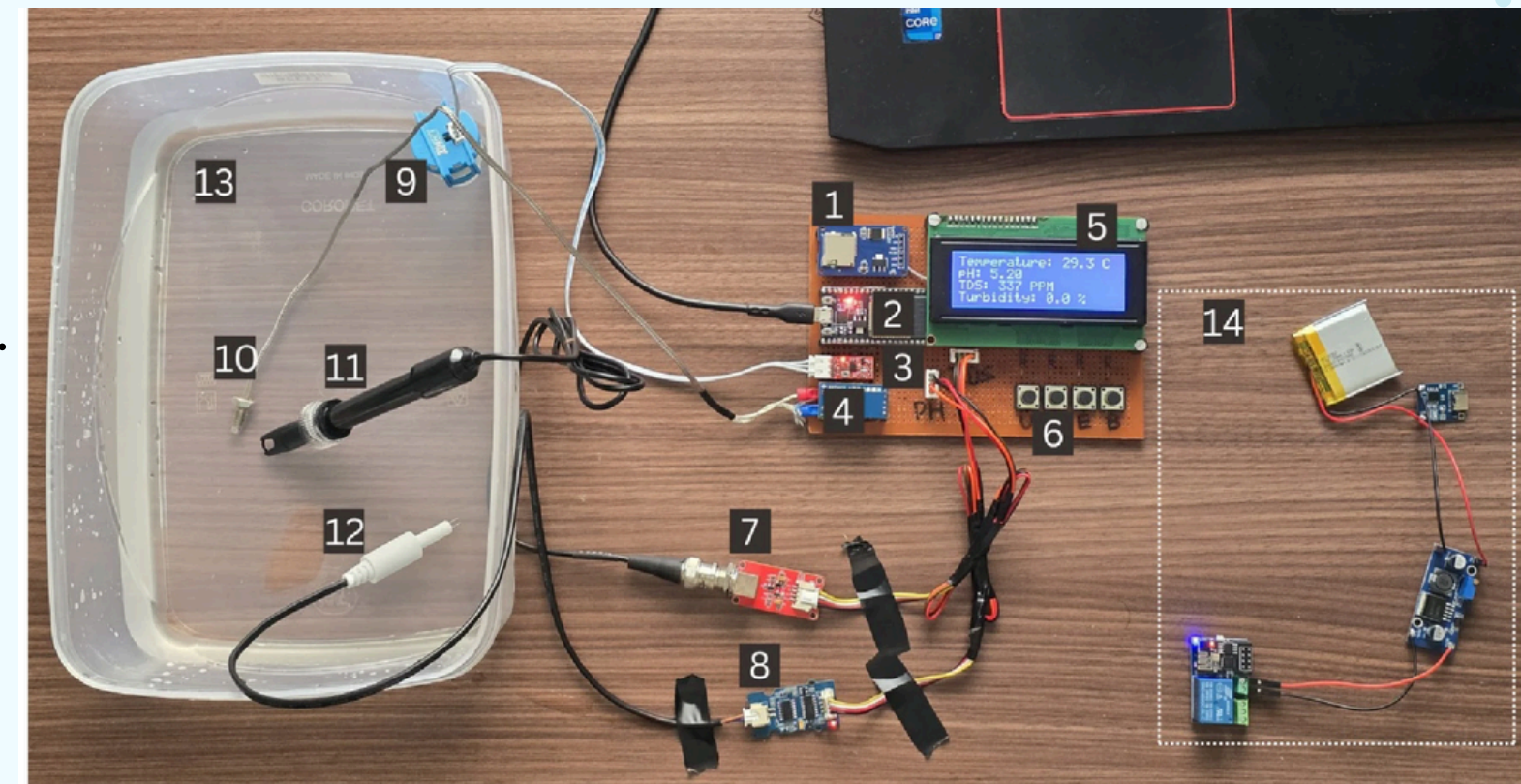


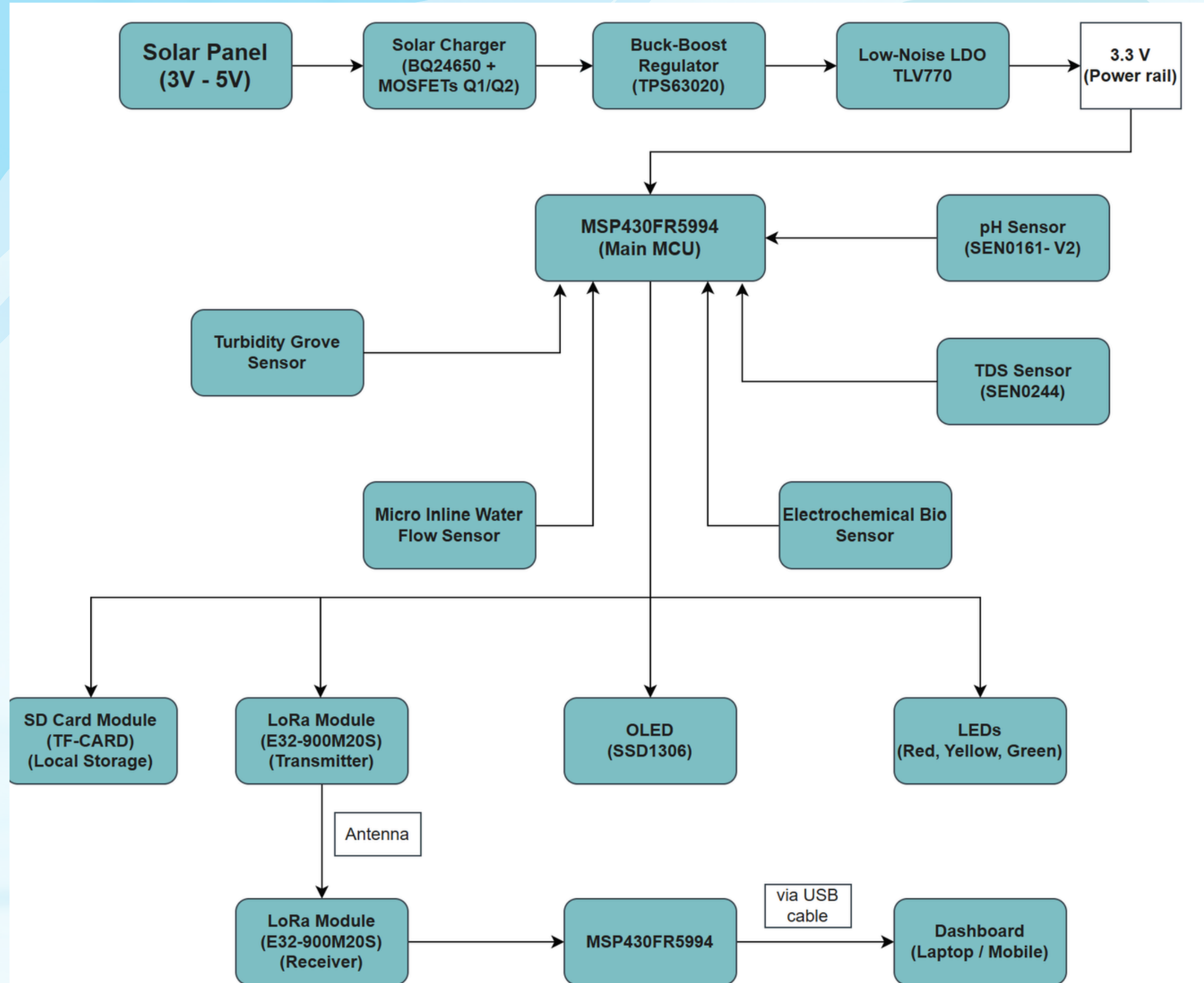
Image reference : “An Intelligent, low-cost water quality monitoring system with on-device machine learning and cloud integration”

How Our Solution Addresses These Issues



Checks Implemented	Concerned Problem	Our Solution
Sensor-based Water Quality Analysis	Contaminated water sources	pH, TDS, turbidity and bio sensors continuously monitor water quality
Automated Threshold Comparison	Delayed detection of unsafe water	Real-time monitoring and instant alerts when parameters exceed safe limits
LoRa-enabled Data Transmission	Lack of transparency	Live dashboard showing water quality status in serial monitor
Multi-Point Flow Measurement	Unequal water distribution	Flow sensors monitor water supply and detect low-flow regions

Block Diagram



Design Thinking Map



Empathizing with the people

- Waterborne diseases (diarrhea, typhoid, cholera, hepatitis) caused 10,738 deaths in India over 5 years and cost 73 million working days lost annually - a \$600 million yearly economic burden. Nearly 70% of all disease outbreaks reported by the Ministry of Health are waterborne.
- Despite JJM deploying 2,843 water testing laboratories and testing 38.78 lakh samples in 2025–26, these are periodic, lab-based checks. There is no real-time, continuous, last-mile monitoring at the point of consumption - the village tap itself.
- Interacted with rural residents to understand their water usage habits and concerns. We found that people often cannot identify water contamination until health problems occur.
- Identified the lack of real-time water quality monitoring at village taps, leading to delayed detection of contamination.

Defining the problem

How might we design a low-cost, solar-powered water quality and equitable distribution monitoring system for rural piped water networks that alerts villagers and authorities to contamination and unfair distribution in real time - using simple visual indicators that require no technical knowledge to understand?

Design Thinking Map



Ideate

Choice of Bluetooth Versus LoRa

Bluetooth was ruled out for the rural deployment due to its limited range. Instead, LoRa was selected because its long-range, low-power capabilities are ideal for village-scale monitoring systems that require reliable data transmission over long distances.

Choice for the number of pressure sensors to detect pipe leak

Initially, a single pressure sensor was considered for leak detection. However, it could incorrectly identify normal pressure fluctuations as leaks, leading to false positives. Therefore, a two-sensor approach was adopted to improve detection accuracy and enable more reliable identification of actual leaks.

Backing up sensor data

Relying solely on live LoRa transmissions risks data loss due to rural power outages, gateway downtime, or signal interference. To ensure data integrity, a local SD card logging mechanism was introduced. By saving time-stamped sensor logs directly on-site, the system guarantees zero data loss during network blind spots and allows authorities to retrieve historical data for offline analysis.

Introducing the concept of weighted average cumulative score

Instead of displaying raw sensor readings, a weighted average cumulative score was introduced to simplify decision making for villagers. Each water quality parameter is assigned a weight based on its relative impact on water safety. The individual contributions are then combined to generate a single score. If the score exceeds a predetermined threshold, the water is classified as safe for drinking; otherwise, an alert is generated.

Design Thinking Map



Quantitative Measurement of Water Quality Score

We normalize each reading onto a strict 0 to 100 scale based on Indian Drinking Water Standards (IS 10500)

Parameter	IS 10500 Reference	Normalization Formula
E. coli	Absent (required)	Detected → 100; Not detected → 0
pH	Target 7.0; Range 5.5–8.5	$\min\left(\frac{ \text{pH} - 7.0 }{1.5} \times 100, 100\right)$
Turbidity	Desirable 1.0 NTU; Max 5.0 NTU	$\min\left(\frac{\text{NTU}}{5.0} \times 100, 100\right)$
TDS	Desirable 500 ppm; Max 2000 ppm	$\min\left(\frac{\text{ppm}}{2000} \times 100, 100\right)$

Thresholds for Red, Yellow and Green Alerts

0 to 25 → GREEN

26 to 50 → YELLOW

50 to 100 → RED

We multiply each measured metric by its assigned weight:

$$\text{Cumulative Weighted Score} = (0.45 \times q_{\text{E.Coli}}) + (0.25 \times q_{\text{pH}}) + (0.20 \times q_{\text{Turbidity}}) + (0.10 \times q_{\text{TDS}})$$

Design Thinking Map



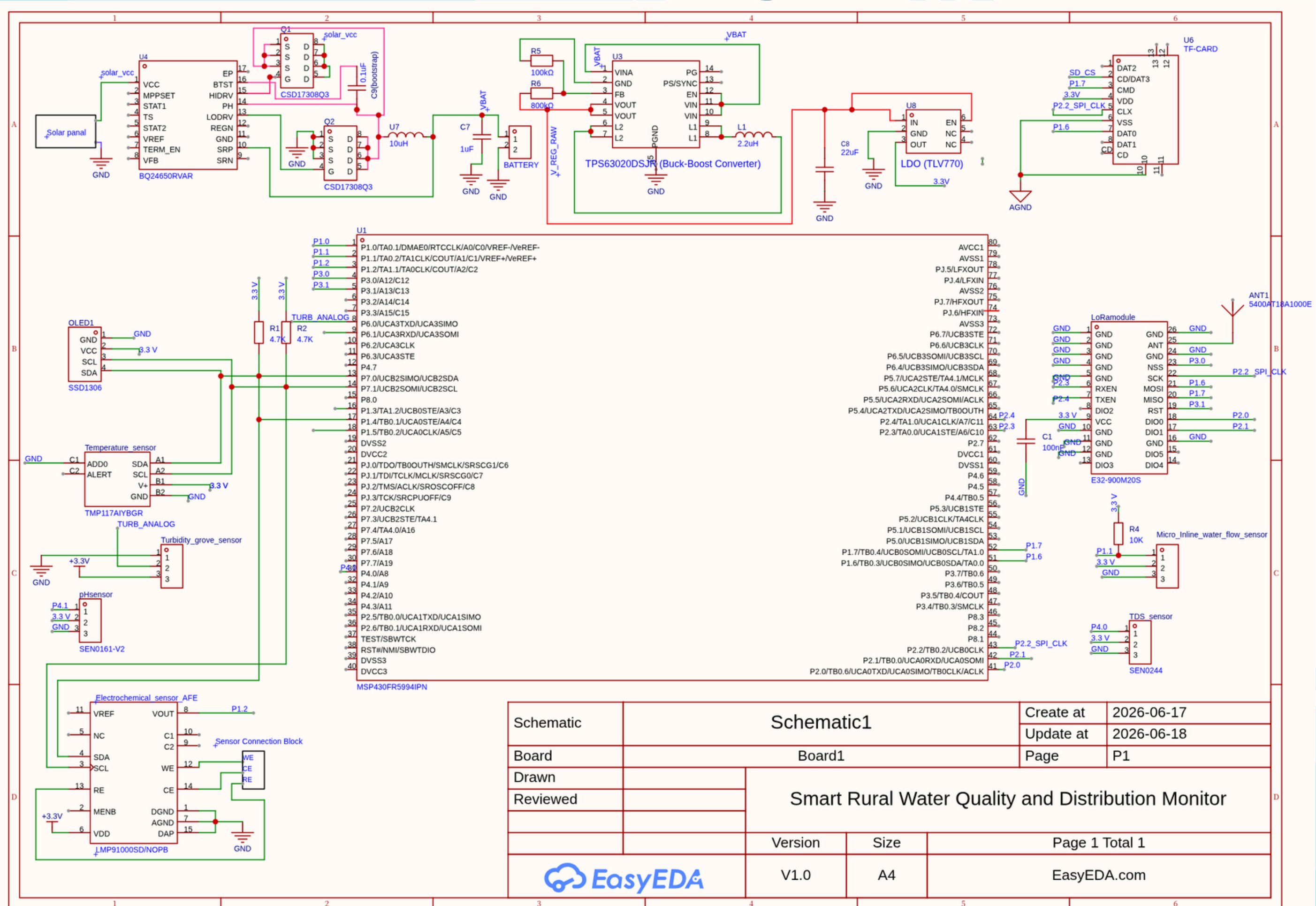
Prototype

- Build a low-cost breadboard model using an MSP430 LaunchPad, pH sensor and turbidity sensor.
- Integrate LoRa communication modules along with SD card logging.
- Implement a color-based alert system (Green = Safe, Yellow = Warning, Red = Unsafe) for easy interpretation.
- Power the prototype using a solar panel and rechargeable battery.

Test

- Test whether users can easily understand the color-based alerts without interpreting scientific values.
- Verify the accuracy of pH and turbidity measurements.
- Evaluate the reliability and range of LoRa communication.
- Assess solar-powered operation and battery backup performance.
- Collect user feedback and refine the system accordingly.

Circuit Diagram



Block level Schematic



Development Boards & Modules

- MSP430 LaunchPad
- pH Sensor Module : SEN0161-V2
- Turbidity Sensor Module : GROVE Turbidity Sensor
- TBS Sensor Module : SEN0244
- E-Coli Sensor Module : LMP91000 + SPCE
- LoRa Communication Module : E32 900M20S
- SD Card Module : TF card
- OLED Display : SSD1306
- Solar Panel
- Charge Controller : BQ24650RVAR
- Buck-Boost Converter : TPS63020DSJR
- LDO : TLV770
- Battery : 3.7 V Li-ion battery
- Flow Rate Module : Micro Inline Water Flow Sensor

Wiring Needed

- Full-size breadboard
- Male-to-male jumper wires
- Male-to-female jumper wires
- 40-pin header strips
- 3-pin header strips
- USB cable for MSP430 programming

TI Parts Used



Category	Name	Purpose
MCU / Processor	MSP430FR5994 (MSP430 LaunchPad)	Processes sensor data and controls the entire system
Analog Front End	LMP91000	Converts low-level electrochemical currents into measurable voltages
LDO	TLV770	Provides a stable, low-noise regulated voltage supply for sensitive analog and digital circuits.
Charge Controller	BQ24650RVAR	Manages solar-powered battery charging using MPPT, and protects the battery during charging.
Mosfets	CSD17308Q3	switches for power-path control, battery charging/discharging, and converter operation.
Buck-Boost Converter	TPS63020DSJR	Regulates the output voltage by stepping it up or down as needed,

Three Success Criteria



Water Quality Detection Accuracy

Target: Accurate identification of safe and unsafe water conditions.

Unfair Water Distribution Detection

Target: Detect abnormal flow patterns and ensure balanced water distribution across different locations.

LoRa Communication Reliability

Target: Successful transmission of sensor readings to the transmission board.

Impact Snapshot



Social Impact

- Early detection of unsafe water conditions and unfair water distribution
- Simple alerts improve community water awareness

Economic Impact

- Reduces cost of frequent manual testing
- Helps prevent losses through faster response

Environmental Impact

- Supports sustainable water resource management
- Solar-powered system reduces energy dependence

Estimated Cost



Component	Quantity	Total Price
MSP430FR5994 LaunchPad	1	₹2500 - ₹2700
pH Sensor Kit (SEN0161-V2)	1	₹3,600 - ₹3,800
TDS Sensor (SEN0244)	1	₹1,100 - ₹1,400
Solar Panel	1	₹3000 - ₹4000
Charge controller	1	₹600
Buck-Boost Converter	1	₹3000
Battery	1	₹500
Turbidity Grove Sensor	1	₹1900 - ₹4000
Inline Water Flow Sensor	1	₹300 - ₹500
Electrochemical Bio Sensor	1	₹2,500 - ₹4,500
OLED (SSD1306)	1	₹150 - ₹500
LoRa module (E32 900M20S)	1	₹450 - ₹700
Miscellaneous costs	-	₹200 - ₹300
Total		₹19,800 - ₹26,500

Future Improvements



ML-BASED PREDICTION

- **Predictive Analysis:** Use historical data to anticipate contamination events.
- **Early Detection:** Spot abnormal trends before they become serious.

GPS-BASED MAPPING

- **Location Tagging:** Tag each monitoring unit with precise location data to quickly identify and isolate affected regions.
- **Faster Response:** Helps authorities quickly locate affected areas and take corrective action.

BLUETOOTH

- **Suitability:** Densely populated urban environments (short-range).
- **Advantage:** Simpler and lower cost than long-range protocols.
- **Contrast:** Bluetooth optimizes short-range urban efficiency, whereas LoRa handles long-range rural coverage.



**THANK
YOU**